

		parametric models.		
	Sections from References:			
Books and References:				
Text Books				
1. Barlow R.E. and Proschan F.(1985). Statistical Theory of Reliability and Life Testing; Ho Rinehart and Winston.				
2. Lawless, J.F. (2003). Statistical Models and Methods for Lifetime (Second Edition), John Wiley Sons Inc., New Jersey.				
References				
3. Bain L.J. and Engelhardt (1991). Statistical Analysis of Reliability and Life Testing Model Marcel Dekker.				
4. Aven, T. and Jensen,U. (1999). Stochastic Models in Reliability, Springer-Verlag, New York, Inc.				
5. Nelson, W (1982). Applied Life Data analysis; John Wiley.				
6. Zacks, S. (1992). Introduction to Reliability Analysis: Probability Models and Statistics Method New York: Springer-Verlag.				

Programme	B. Sc. Statistics				
Course Code	STA6EJ303				
Course Title	Life Time Data Analysis				
Type of Course	Major Elective				
Semester	VI				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites					
Course	The student has a thorough knowledge of the basic theory of stochastic				

Summary Objective	modelling and statistical analysis of survival data, including graphical techniques. This includes both parametric and non-parametric analysis of censored survival data and data for recurrent events, as well as related regression models
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Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the basic concepts and ideas of survival analysis.	U	C	Instructor-created exams / Quiz
CO2	Examine the properties and methods for standard survival time distributions	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Estimate survival functions using parametric and non-parametric methods.	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Apply and interpret semi-parametric and parametric regression models for survival data.	U	C	Instructor-created exams / Home Assignments
CO5	To apply the concepts learned in the previous modules to a real-life data set.	Ap	P	One Minute Reflection Writing assignments
CO6		Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I			10	15
	1	Basic Quantities and Models- Survival Function, Hazard function, Mean residual life function		
	2	Common Parametric models for survival data.		

	3	Log location scale models,		
	4	Mixture models.		
	Sections from References:			
II			10	15
	5	Right censoring		
	6	Left censoring		
	7	Interval censoring		
	8	Truncation		
	9	Likelihood construction for censored and truncated data.		
	Sections from References:			
III			18	20
	10	Nonparametric Estimation of Basic Quantities		
	11	Estimators of the Survival Functions for Right-Censored Data		
	12	Estimators of Cumulative Hazard Functions for Right-Censored Data		
	13	Point-wise Confidence Intervals for the Survival Function		
	14	Life Table		
	15	Estimation of Survival in the Cohort Life Table.		
	16	Hypothesis testing- One sample tests		
	17	Tests for two or more samples.		
	Sections from References:			
IV			10	20
	18	Semiparametric Proportional Hazards Regression with Fixed Covariates		
	19	Model Building Using the Proportional Hazards Model		
	20	Graphical Checks of the Proportional Hazards Assumption,		
	21	Additive hazards regression models.		
	22	Regression Diagnostics		
	Sections from References:			
V	Open Ended Module:		12	
	1	Practical exercises on lifetime data using the statistical software R:		